

Make systems legible to humans and to AI.

HMS is three products on one semantic foundation: **DataPro**, **DataMCP**, and **DataViz** — a unified, deterministic, queryable view of your entire stack.

THE PROBLEM

Companies don't have an observability problem. They have a **context** problem.

Engineers piece together what's happening from a dozen disconnected places — Datadog, Grafana, Databricks, Jupyter, log files, custom UIs, Slack, runbooks, repos.

01

Knowledge is scattered

Each tool sees one slice. Nobody sees the system as a whole.

02

Onboarding takes years

Building intuition for a complex system takes hundreds of hours per engineer — and never compounds across teams.

03

Incidents take hours

"What is this service? Who owns it? Has this happened before?" Should be seconds. Are routinely hours.

AI agents inherit the same chaos — and they can't reason their way out.

Every AI agent built for ops, incident response, or developer productivity hits the same wall: it doesn't actually **understand** the system it's operating on.

TODAY'S AI AGENTS

Stitching, guessing, hallucinating

- Call random APIs and databases directly
- Generate brittle SQL and PromQL on the fly
- Guess relationships that aren't documented
- Hallucinate connections between services
- No memory of what's true vs. inferred

WHAT THEY NEED

A trustworthy semantic layer

- Ask for an entity by name, get the truth
- Follow named relationships, not column joins
- Field-level provenance, not raw values
- Bounded, predictable query behavior
- One API, every system in the company

THE INSIGHT

Make systems legible.

Both humans and AI struggle with the same thing: the data exists, but it isn't structured, connected, or trustworthy. Build that layer once — and everyone wins.

THE SOLUTION

HMS is three products. One semantic foundation.

Each product serves a different consumer. They share the same core, so adding one strengthens the others.

FOUNDATION · PHASE 1

DataPro

The semantic data layer. Turns scattered sources into typed objects with identity, relationships, and provenance. Deterministic runtime. The foundation everything else stands on.

AGENT INTERFACE · PHASE 2

DataMCP

The AI agent surface. Exposes DataPro through the Model Context Protocol so AI agents can ask structured questions about systems — and get reliable, bounded, provenance-tracked answers.

HUMAN INTERFACE · PHASE 3

DataViz

The 2D spatial dashboard. Lets engineers pan, zoom, and explore their entire system as a navigable map — building intuition the way humans actually think about space.

One question. Two worlds.

"Why is checkout-pricing failing?"

3:14 AM

WITHOUT HMS

Page fires. Open Datadog. Switch to Grafana. Open the K8s dashboard, then the deploy log, then the monorepo. Search Slack. Ping the owner — who's asleep. Check the incident channel from last week. Skim a runbook that's two years out of date. Cross-reference three timestamps in three different timezones. The fix turns out to be a one-line config flag from a deploy 8 hours earlier.

TIME TO RESOLVE: 47 MINUTES

3:14 AM

WITH HMS

Page fires. The on-call AI copilot calls **DataMCP** for the **checkout-pricing** object. **DataPro** assembles a single resolved view: current status, last 3 deployments, dependencies, recent error metrics, and the engineer who shipped the change. The copilot proposes the rollback. The human reviews it in **DataViz** and clicks approve.

TIME TO RESOLVE: 90 SECONDS

The 3am story is one of three reasons companies adopt HMS.

Defense first. Offense follows.

DEFENSE I · INSTITUTIONAL MEMORY

The senior engineer leaves. The system still knows what they knew.

Every connector configuration, link declaration, identity rule, and safety contract is captured as the side effect of using HMS — durable, reviewable, version-controlled. The new hire's first day starts with a working system and an AI agent that can answer questions about it, not a treasure hunt across six tools and three departed teammates.

Mapped to **DataPro** — the configurations are the institutional record.

DEFENSE II · INCIDENT RESPONSE

At 3am you ask one question — and HMS answers it.

The user story you just saw. In an incident, the bottleneck is reach, not data. HMS replaces three engineers in three Slack channels writing their own ad-hoc SQL with one conversational, visual surface. Safety contracts mean even broad questions don't take down the source systems.

Mapped to **DataMCP + DataViz** on the **DataPro** substrate.

OFFENSE · COMPLEXITY MAP

Adding a service is a contained operation, not a tax on the whole org.

Complexity tax compounds — every new service adds N "how does this relate to..." conversations, and N grows superlinearly. With HMS, adding a service is one connector + a few link declarations. The link store becomes your company's complexity map: explicit, queryable, evolvable. Also the substrate AI agents need to reason about the system.

Mapped to **all three products** — especially the link layer.

What a DataPro query actually looks like.

QUERY

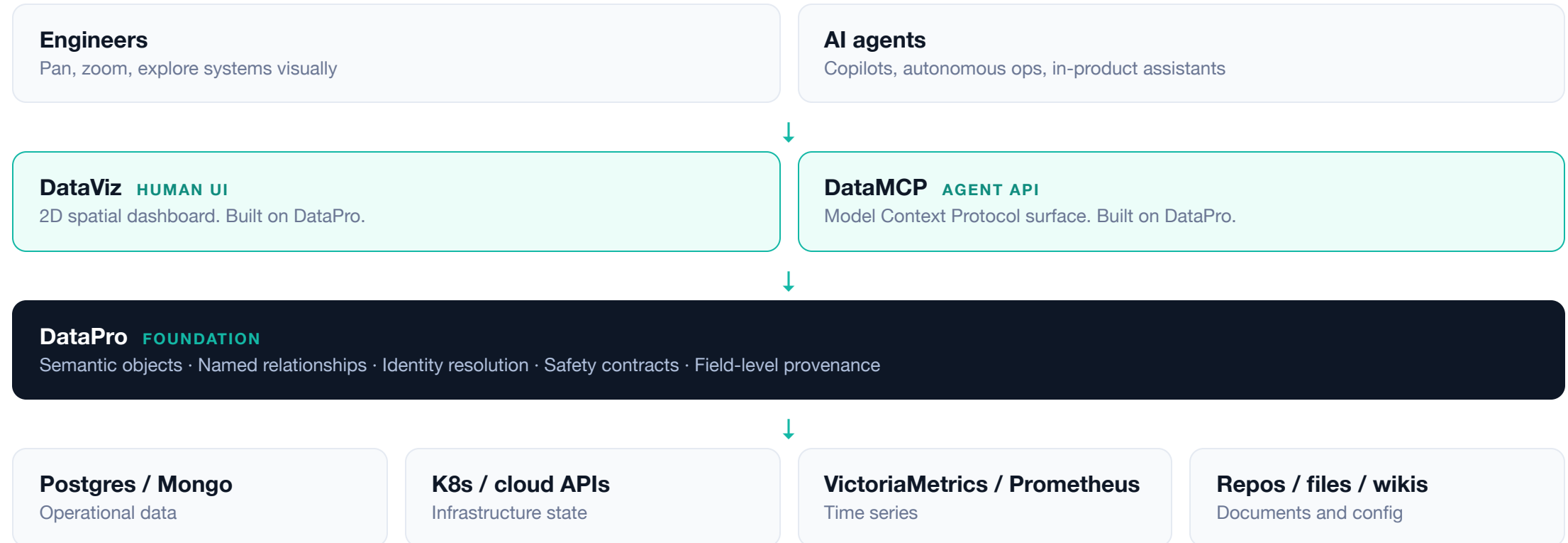
```
// One question, no SQL, no PromQL
{
  "from": "AppInstance",
  "where": { "name": "checkout-pricing" },
  "include": {
    "host": true,
    "owner": true,
    "recent_deployments": {
      "within": "24h"
    },
  },
  "metrics": {
    "names": ["error_rate", "latency_p99"],
    "within": "1h"
  }
}
```

RESPONSE — ASSEMBLED FROM 4 BACKENDS

```
{
  "type": "AppInstance",
  "id": "prod/checkout-pricing",
  "status": "degraded",           // from K8s
  "owner": "payments-team",      // from Postgres
  "host": { "id": "h-918", "region": "us-east-1" },
  "recent_deployments": [
    { "sha": "a3f1e9", "at": "2026-04-28T19:02Z",
      "author": "jane@..." }    // from Git
  ],
  "metrics": {                   // from VictoriaMetrics
    "error_rate": 0.12,
    "latency_p99_ms": 2840
  },
  "_provenance": { "all_fields": "ok" }
}
```

The hard part isn't the syntax — it's resolving identity across sources, planning the joins via a link store, enforcing query safety contracts, and assembling a provenance-tracked result. That's DataPro. **DataMCP** exposes this same shape over MCP for AI agents; **DataViz** renders it spatially for humans.

Three products, layered.



WHAT IT UNLOCKS

One layer. Many products on top.

VIA DATAMCP

Incident response copilots

"Why is this service failing?" Agents traverse dependencies, recent deployments, and metrics through one API — instead of calling six tools and guessing.

VIA DATAVIZ

Engineering onboarding

New engineers explore their company's systems through a 2D spatial map — building intuition in days, not months.

VIA DATAPRO

Custom internal tools

Teams build the dashboards they actually need against one consistent backend — no more bespoke data fetchers per tool.

VIA DATAMCP

Autonomous ops agents

Long-running agents that monitor, investigate, and act — anchored to a single, deterministic source of truth.

Everyone else owns a slice. HMS owns the layer above them.

CATEGORY	EXAMPLE	WHAT THEY DON'T DO
Telemetry & dashboards	Datadog, Grafana, New Relic	No semantic model. No cross-source object identity. AI-hostile.
SQL federation engines	Trino, Presto, Starburst	Raw SQL surface. No object model, no link semantics, no safety contracts.
Service catalogs	Backstage, Cortex	Static metadata only. Don't query live data, no AI-native interface.
Bespoke internal tools	Every large eng org	The fragmented status quo HMS replaces.
HMS	DataPro · DataMCP · DataViz	Semantic objects · Identity resolution · Link store · Safety contracts · MCP-native by design

Three timing windows, all open at once.

2026 — AI AGENTS

The agent ecosystem is real

MCP, tool-use, multi-step agents — all maturing fast. They desperately need structured, trustworthy context.

FATIGUE

Observability is broken

CTOs are openly tired of stitching five vendors to answer one question. The market wants one layer above all of them.

FOUNDATIONS

The substrate finally exists

Trino, MCP, modern agent frameworks — heavy infra is now off-the-shelf. We focus on the semantic layer.

MARKET

A wedge into three converging markets.

TAM

\$85B

Observability + DevOps + AI infra

Observability (\$60B by 2028), internal developer platforms, and emerging AI agent infrastructure — all growing double-digit annually.

SAM

\$8B

Mid-market & enterprise eng orgs

~30K companies globally with 50+ engineers and material multi-source data sprawl.
Average willingness-to-pay: \$30–500K ACV.

SOM (YEAR 3)

\$120M

AI-forward eng teams

Initial wedge: companies actively building AI agents on their internal systems.
Underserved, fast-moving, willing to pay early.

BUSINESS MODEL

Two on-ramps, one platform.

PRIMARY · CLOUD SAAS

Per-source & per-seat pricing

Hosted HMS Core + Dashboard. Tiered by number of connected data sources, query volume, and engineer seats. Self-serve for small teams, sales-assisted for enterprise.

TARGET ACV · \$30K–\$120K

ENTERPRISE · SELF-HOSTED

Annual license + support

For regulated and large enterprise customers running HMS on their own infrastructure. Includes premium connectors, identity integration, and dedicated support.

TARGET ACV · \$150K–\$500K+

An open-source HMS Core seeds bottom-up adoption inside engineering teams; commercial features (Dashboard, AI agents, premium connectors, multi-tenant identity, SLAs) drive revenue.

ROADMAP

A thin wedge that compounds.

Each phase ships a real product. Each one makes the next more valuable.

PHASE 1 — NOW

DataPro

The deterministic semantic data layer. One query, multi-source results. Tools and design partners build on the API.

PHASE 2 — NEXT

DataMCP

MCP interface for AI agents. Cleanest early demo: an AI copilot debugging incidents through HMS.

PHASE 3

DataViz

2D spatial dashboard built on DataPro. Maps you can pan, zoom, and remember.

PHASE 4

Ecosystem

Marketplace of connectors, object types, and link packs across all three products. Network effects compound.

TEAM & ASK

What we've done. What we need.

The team

[Founder name] — [Background: prior role, notable systems shipped, why-this-why-now credibility].

[Co-founder name] — [Background, complementary expertise, prior collaboration with founder].

Advisors — [List, especially anyone with infra, observability, or AI-agent credibility].

*Replace this slide with your real team narrative — focus on why **this** team is uniquely suited to build HMS.*

THE ASK

\$3M

Seed round · 18 months runway

- Ship DataPro + first 4 connectors
- Launch DataMCP for the agent ecosystem
- 10 design-partner deployments
- First \$500K ARR